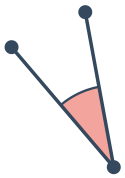
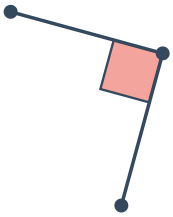
Review: Angles and angle measure

1 State whether the following angles are acute, obtuse, right or straight.

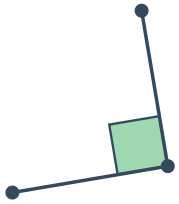
a



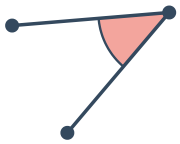
b



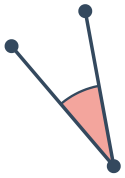
c



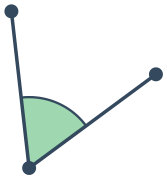
d



e



f



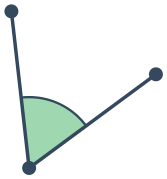
g



h



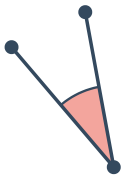
i



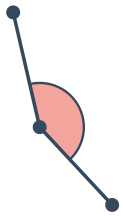
j



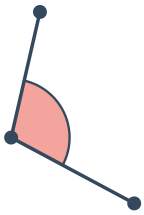
k



l

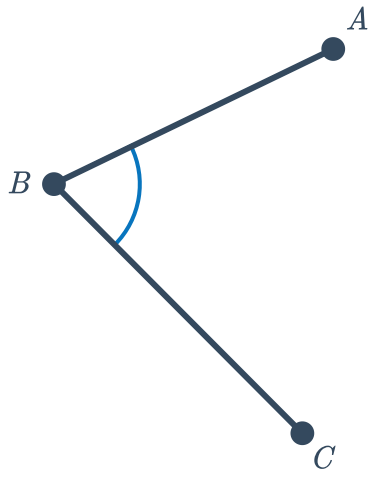


m



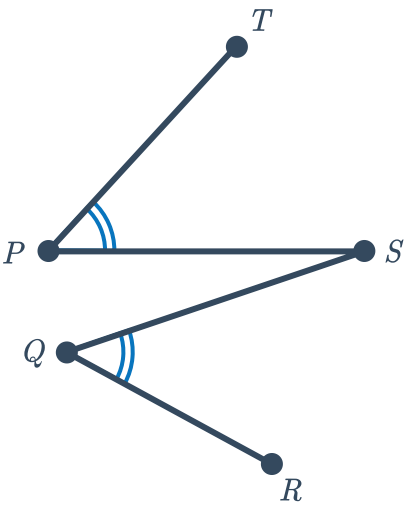
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2 This diagram has the angle $\angle ABC$ marked. Write another way of referring to the same angle.

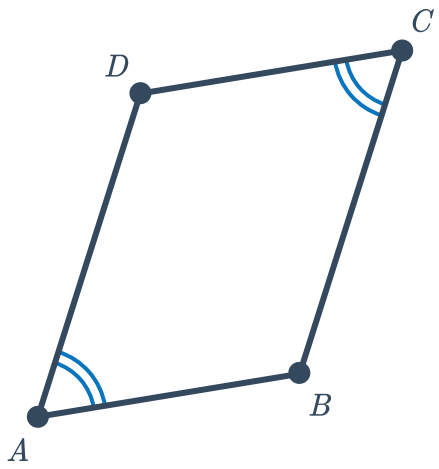


3 For the following diagrams, write the two angles that have been marked congruent.

a



b



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4 State the diagram where $\angle BAD$ and $\angle DCB$ are marked congruent:

Diagram 1

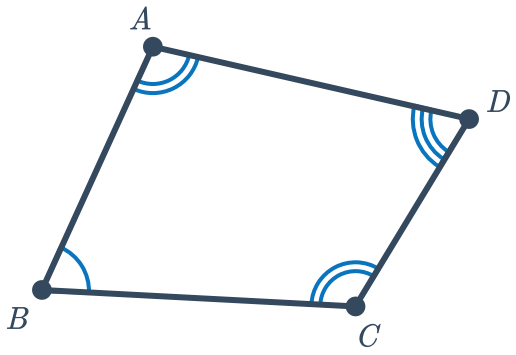


Diagram 2

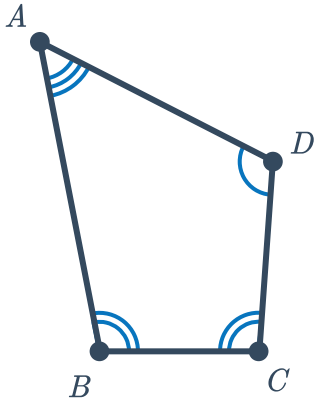


Diagram 3

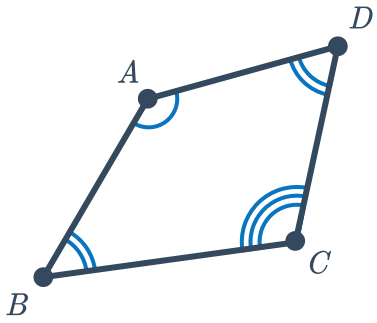
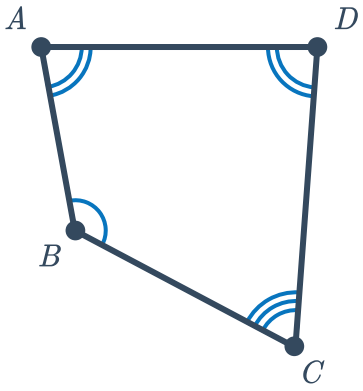
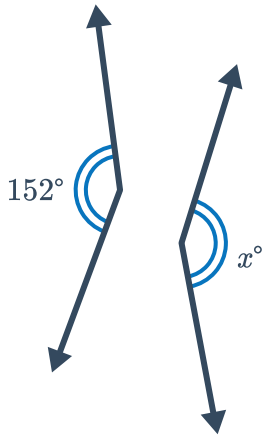


Diagram 4

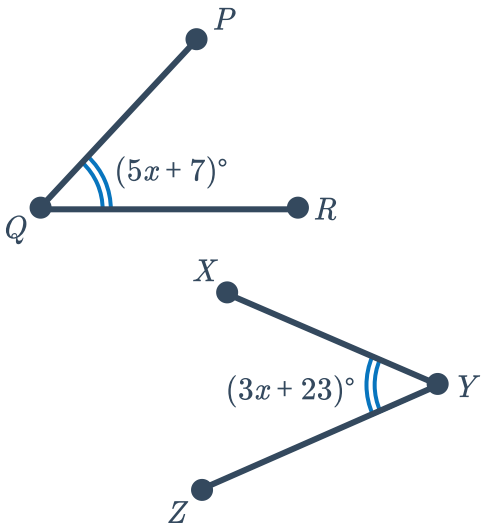


5 Consider the given diagram:
Solve for x.



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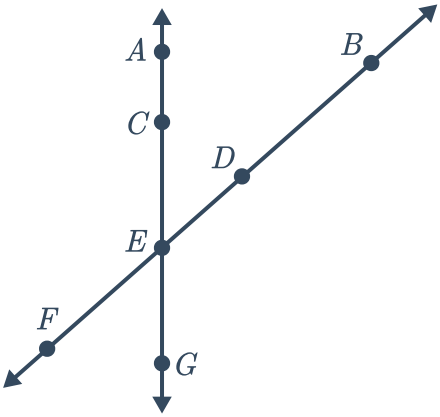
6 Consider the angles in the given diagram:



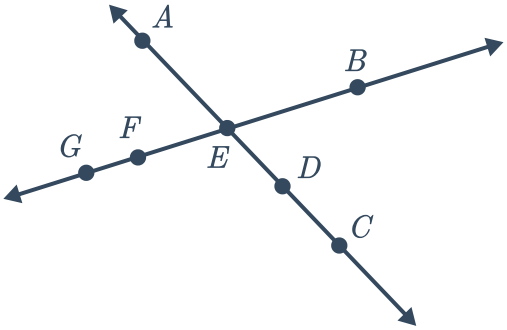
a Write an equation and solve for x.

b Hence, find $m\angle PQR$.

7 For the given diagram, write the rays that contain the same points as $\overrightarrow{FD}$.



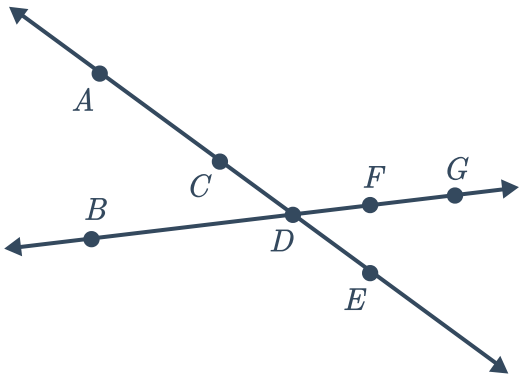
8 For the given diagram, write the rays that contain the same points as $\overleftrightarrow{AC}$.



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9 For the given diagram, write the pair of opposite rays.

- $\overrightarrow{DG}$
- $\overrightarrow{DA}$
- $\overrightarrow{AD}$
- $\overrightarrow{DE}$



10 For the following angles:

i Write the point that is the vertex of the angle.

ii Write the two rays that form the legs of the angle.

a $\angle RST$

b $\angle PQR$

11 Classify $\angle ABC$ as obtuse, acute, straight or right if $m\angle ABC$ is:

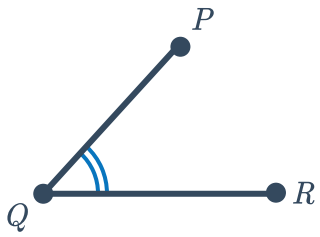
a 25° **b** 64° **c** 90° **d** 143°

e 113° **f** 180°

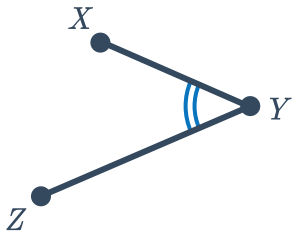
Additional questions

12 Name each angle shown in three different ways.

a



b



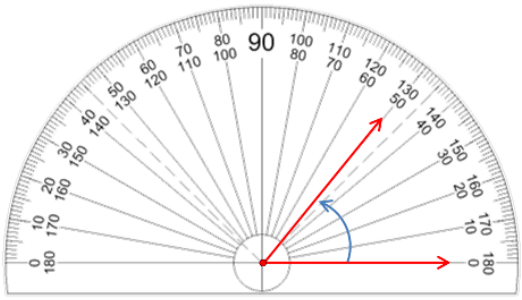
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13 For each angle shown

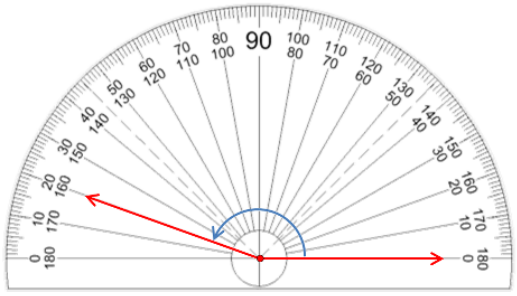
i Identify the angle being measured

ii Use the measure to classify the angle as acute, obtuse, right, or straight

a

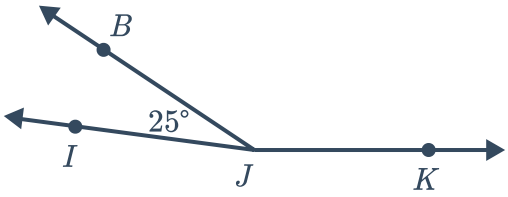


b

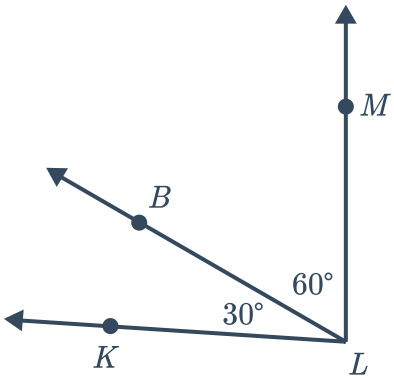


14 Find the measure of each angle

a If $m\angle IJK = 150^\circ$, find $m\angle BJK$

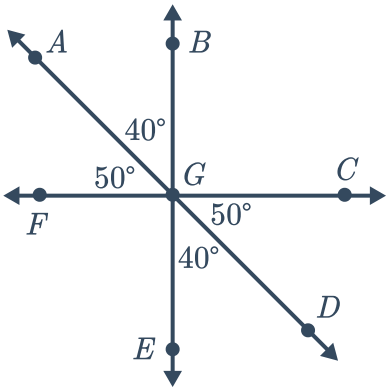


b Find $m\angle KLM$

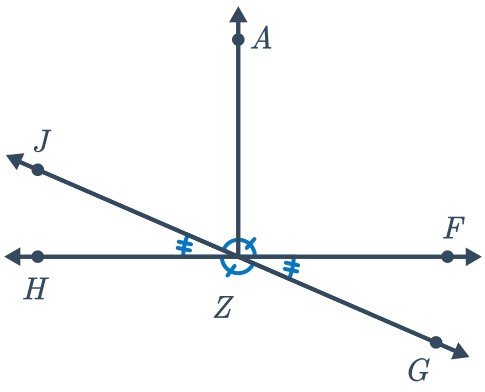


15 Use the diagrams to identify two pairs of congruent angles.

a

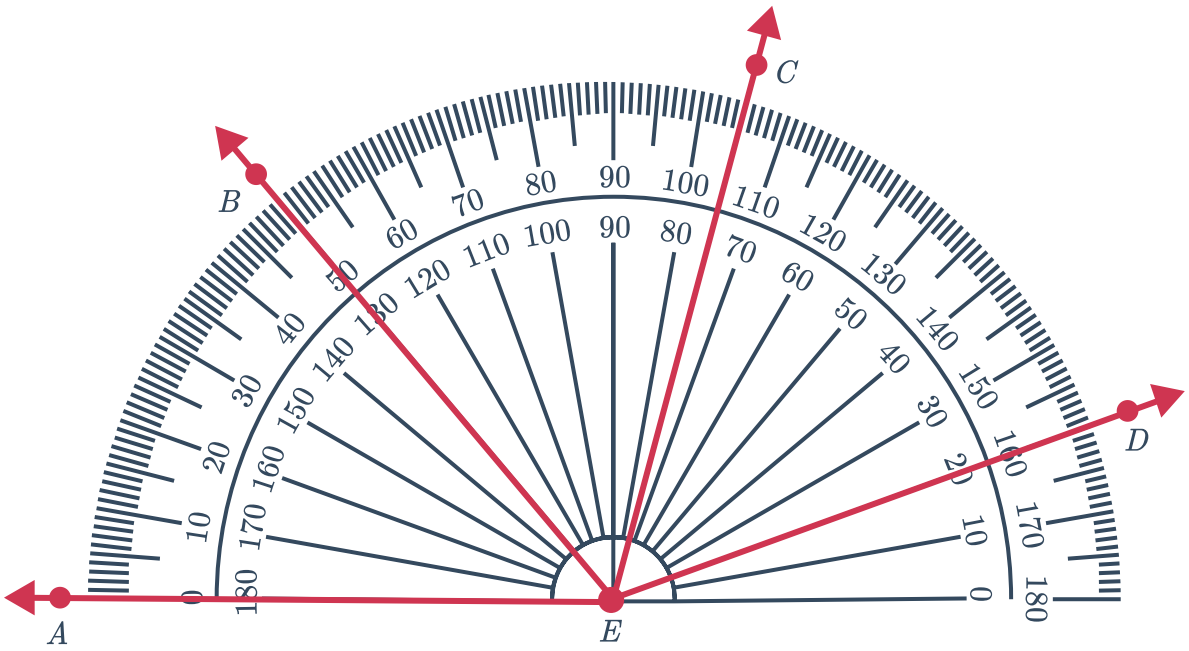


b



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16 Consider the following figure



i Use the diagram to find each angle measure.

ii Classify the angle as acute, obtuse, right, or straight

a $m\angle AEB$

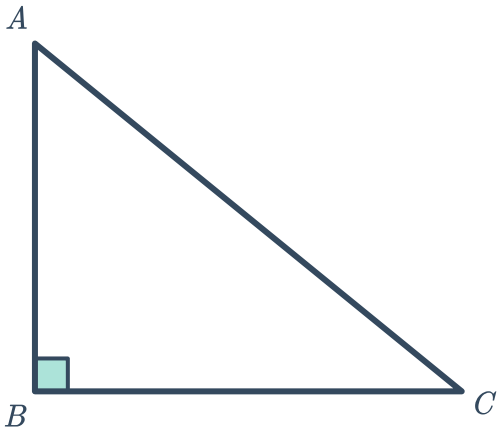
b $m\angle AED$

c $m\angle CED$

d $m\angle DEB$

17 Each statement shown has an error. Identify the error and rewrite the statement correctly.

a



In the triangle shown $\angle ABC = 90^\circ$

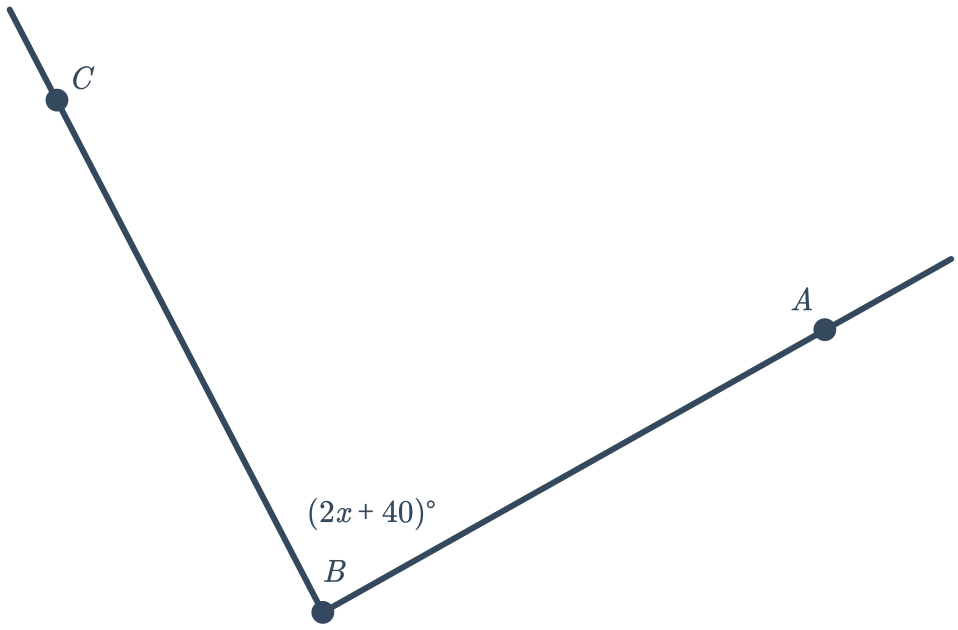
b In $\angle XZY$, Y is the vertex.

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c 180° is an obtuse angle since its measure is greater than 90°.

18 When one ray of an angle aligns up with 10° the other ray aligns with 100°. If the protractor is moved so that the first ray of the angle now aligns with 35° what value will the second ray align with?

19 Josiah wants to find the value of x. Explain the mistake Josiah made.



Josiah's work:

$$2x + 40 = 90$$

$$2x = 50 \quad \text{Subtract } 40$$

$$x = 25 \quad \text{Divide by } 2$$

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